Bank Marketing Campaign Analysis

Exploratory Data Analysis Project

1. Introduction

This project examines the UCI Bank Marketing dataset to understand which customer characteristics and campaign-related factors are associated with term-deposit subscriptions.

2. Dataset Overview

The dataset contains 45,211 records and 17 original columns. The target variable, y, records whether a customer subscribed to a term deposit.

3. Data Preparation

The initial inspection found no missing values reported by Pandas and no exact duplicate rows. Some categorical fields contain the value unknown, which is distinct from a missing value and should be considered during interpretation.

The target was converted into a numerical indicator: 1 for a successful subscription and 0 otherwise.

4. Key Findings

There were 5,289 successful subscriptions out of 45,211 records, giving an overall subscription rate of 11.70%.

Students had an observed subscription rate of 28.68%, and retired customers 22.79%. The rate for tertiary education was 15.01%.

Cellular contacts had a subscription rate of 14.92%, compared with 13.42% for telephone contacts and 4.07% for records with an unknown contact method. March had the highest observed monthly subscription rate at 51.99%.

Successful subscriptions tended to have longer call durations in the visual comparison.

5. Interpretation

Subscription rates vary across job groups, education categories, contact methods and months. These are associations in the observed dataset; they do not prove that any single characteristic caused a subscription.

Monthly rates can be influenced by differences in campaign timing, customer composition and sample sizes. High observed rates should be examined alongside the number of records in each group.

Call duration is measured after a call takes place, so it should not be treated as information available before contacting a customer.

6. Recommendations

1. Investigate the needs and preferences of student and retired customer groups.

2. Compare campaign outcomes across contact methods while considering group sizes and costs.

3. Examine monthly campaign performance before deciding when to allocate future resources.

4. Review previous campaign outcomes to identify potentially useful customer segments.

5. Evaluate contact frequency carefully rather than assuming that more calls produce better results.

7. Limitations

The analysis is observational and does not establish causation. Some categorical values are recorded as unknown. The findings reflect this dataset and may not generalize to other campaigns or time periods.

8. Conclusion

The analysis identifies meaningful differences in subscription rates across customer segments and campaign characteristics. These patterns can guide further investigation and help improve future campaign planning.

Dataset source: UCI Machine Learning Repository, Bank Marketing. Report values are based on the results provided during the analysis workflow; verify them against the saved notebook before final submission.